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section: 02

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Lecture: 5

Practice problem set 2

Q:1

(a) $M \times N = 4 \times 4$ [input] $\quad m \times n = 3 \times 3$ [Kernal]

$P = ?$ $\quad H \times W = 4 \times 4$ [given]

$$H = \frac{M + 2P - m}{S} + 1$$

$$\Rightarrow 4 = \frac{4 + 2P - 3}{1} + 1$$

$$\Rightarrow P = 1$$

(b)

6	2	1	9
2	0	2	4
1	9	9	5
2	0	1	2

$\frac{1}{16}$

1	2	1
2	4	2
1	2	1

pixel values:

$$\frac{1}{16}(6\times1)+(2\times2)+(1\times1)+(2\times2)+(0\times4)+(2\times2)+(1\times1)+(9\times2)+(9\times1)$$

$$= \frac{47}{16} = 2.9 \approx 3$$

$$\frac{1}{16}(2+2+9+0+8+8+9+18+5)$$

$$= 3.8 \approx 4$$

$$\frac{1}{16}(2+0+1+2+36+18+2+0+1)$$

$$= 3.9 \approx 4$$

$$\frac{1}{16}(0+4+4+18+36+10+0+2+2)$$

$$= 4.75 \approx 5$$

output blurred:

3	4
4	5

(c) upsharp Masking = Original $f(x,y)$ − blur $f(x,y)$

cropped Original Img:

$$\begin{bmatrix} 6 & 2 \\ 2 & 0 \end{bmatrix}$$

from 'b'

blurred Img:

$$\begin{bmatrix} 3 & 4 \\ 4 & 5 \end{bmatrix}$$

upsharp mask = $\begin{bmatrix} (6-3) & 2-(4) \\ 2-4 & 0-5 \end{bmatrix} \rightarrow \begin{bmatrix} 3 & -2 \\ -2 & -5 \end{bmatrix}$

(d) Given, $K=1$, upsharp masking = Original Img + K × Upsharp Mask

$$= \begin{bmatrix} 6 & 2 \\ 2 & 0 \end{bmatrix} + 1 \times \begin{bmatrix} 3 & -2 \\ -2 & -5 \end{bmatrix} = \begin{bmatrix} 9 & 2+(-2) \\ 2+(-2) & 0+(-5) \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 0 \\ 0 & -5 \end{bmatrix}$$

(Ans)

(e) $K=2$, if $K>1$ then it's a high boost filtering :

high boost filtering Img = original Img + 2 × Unsharp mask

$$= \begin{bmatrix} 6 & 2 \\ 2 & 0 \end{bmatrix} + 2\begin{bmatrix} 3 & -2 \\ -2 & -5 \end{bmatrix} \Rightarrow \begin{bmatrix} 6 & 2 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 6 & -4 \\ -4 & -10 \end{bmatrix}$$

$$= \begin{bmatrix} 12 & -2 \\ -2 & -10 \end{bmatrix}$$

(Ans)

Q:2 Sobel operator.

Q:3 Robinson Compass Masks.

Q:4 (a) Salt & pepper noise.

(b) Median filter which is non-linear & effective in removing salt & pepper noise without blurring edges.

(c) Detecting soft edeges a 2D kernel which is Laplacian operator:

(+ve)
$$\begin{matrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{matrix}$$

(-ve)
$$\begin{matrix} 0 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 0 \end{matrix}$$

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(d) Yes, by applying unsharp masking.

(e) NO, it's already blurred. Blurred Img for sharping is not effective.

Lecture: 6

Ex: 1 In Image 1 all values are same, the image contain a low-frequency component. DFT result will have a strong DC at F(0,0) & other frequency component will 0. So, it represents a smooth region. Image 1 reprents a blur image & can be linked to low-pass filtering.

Image 2 contain high-frequency variations along the horizontal axis. The DFT result will show energy concentered at higher frequency, indicating strong edeges or transitions in the image. So, image-2 resembles the effect of high pass fitering.

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Ex: 2 → D

Ex: 3 → A

Ex: 4 → B

Ex: 5 → Image 1 → 2nd DFT [less diff]

Image 2 → 1st DFT [high/huge diff]

Q: 6 ~~Ex~~ We will use high pass filter for img 1 it suppresses low-frequency componets & retain high frequency components. sharpend Img with enchaned edeges,

$$H(u,v) = \begin{cases} 0 & \text{if } D(u,v) \le D0 \\ 1 & \text{if } D(u,v) > D0 \end{cases}$$

We will use low-pass filter. Because it suppresses high frequency components & retrain low frequency components. Result is a blurred or smooth Image.

$$H(u,v) = \begin{cases} 1 & \text{if } D(u,v) \le D0 \\ 0 & \text{if } D(u,v) > D0. \end{cases}$$

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